

Information Flow, Volatility and Fat-tailed Distribution of Cryptocurrencies: An Empirical Analysis

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This project is an outgrowth from a discussion with Prof. J. Huston McCulloch at the Ohio State University about the degree of freedom for goodness-of-fit tests for Paretian stable distribution and Merton's (1976) compound Poisson jump model. The authors also benefit from a follow-up discussion with Prof. Robert Merton when he visited the University of Macau. The authors are solely responsible for any errors in this paper.

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Abstract

This study investigates whether the nonnormality of unconditional return distribution is due to volatility persistence created by random arrival of information. Employing the stable-GARCH model, it examines the information impact on the return and volatility of cryptocurrencies after adjusting for heteroskedasticity. The findings show that information flow significantly and positively affects volatility, while negatively influences tail heaviness of cryptocurrency return distribution. After considering the random arrival of information within trading days, the return distribution is still heavy tailed in calendar days. It provides practical implications for diversification, asset pricing and risk management in cryptocurrency markets.

Keywords: Information Flow, Cryptocurrency, Fat-tailed Distribution, Volatility, Stable-GARCH

JEL classification: C46, C58, G14, G32

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1. Introduction

Cryptocurrencies entered a new era after the U.S. establish the Strategic Crypto Reserve and pass the GENIUS Act in 2025. Cryptocurrency can be highly volatile (see Figure 1), as documented by Bakas et al. (2022) and Eska et al. (2024). Being a reserve asset, the substantial volatility of cryptos may challenge financial stability. Notably, the return distribution of cryptos exhibits heavy tails, conditional heteroskedasticity, and volatility clustering, indicating a higher probability of extreme gains or losses compared to the normal assumption, and such excessive fluctuation could be persistent (Nikolova et al., 2020; Tran and Kukal, 2022; Malek et al., 2023). These characteristics highlight the importance of understanding and monitoring associated risks of cryptocurrencies (Demir et al., 2018; Corbet et al., 2020).

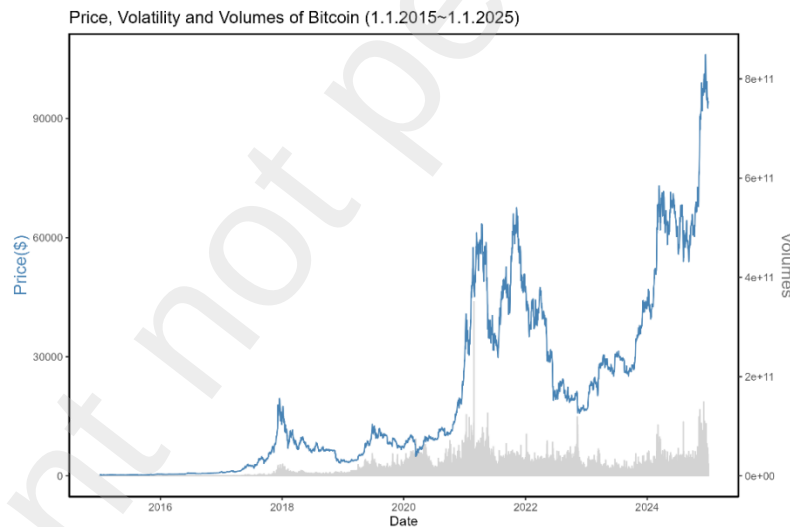


Fig. 1. Price, Volatility and Volumes of Bitcoin

Many studies attempt to explain the volatility of cryptocurrencies. Eska et al. (2024) find that volatility is associated with crypto designs such as age, transaction fees and Proof-of-Stake protocols. Wang et al. (2023) show blockchain features, e.g. average block size and hash rate, contribute to cryptocurrency volatility. Bakas et al. (2022) suggest bitcoin volatility is driven by factors including Google trends, total circulation, consumer confidence and the S&P500 index. However, the explanation of extreme price fluctuation

of cryptocurrency from the perspective of information flow theories is limited. Almeida et al. (2024), Assaf et al. (2023), Caferra (2022), Zhou et al. (2024) and Sheraz et al. (2022) use entropy-based methods to analyze information flow of cryptos. But research gaps exist in linking economic theories to the presence of heavy tailed return distribution and time-varying volatility of cryptocurrencies.

The shape of return distribution is influenced by information affecting the intrinsic value and how that information flows to the market (Damodaran, 1985). Fama's (1965b, 1970) Efficient Market Hypothesis (EMH) states that asset prices fully and instantly reflect all available information, hence no one can earn arbitrage profits. On the other hand, Capital Asset Pricing model (CAPM) and the Option Pricing model (OPM) emphasize the importance of diversification and hedging mechanisms in managing risk. Whether arbitrage profits can be made from cryptocurrency markets¹ is affected by information flow and its impact on prices, i.e., the statistical behavior of the price changes. The subordinated distribution introduced by Clark (1973) showed that price changes over "calendar time" reflect an accumulation of new information during a particular "trading time". Due to the randomness of information arrival, the unconditional return distribution presents fat-tails within "calendar time". It is a subordinated stochastic process with a mixture of normal distributions, in which information flow is the directing process. After adjusting for the random arrival of information, the logarithm of returns is suggested to follow normal distribution (hereafter log-normal) by Clark (1973), while stable distribution (hereafter log-stable) is recommended by Mandelbrot and Taylor (1967), Granger and Morgenstern (1970) and Mandelbrot (1997). Conflicting results remain in terms of the return and volatility patterns accounting for information flow.

Moreover, subordination provides a theoretical foundation for the innovation term in generalized autoregressive conditional heteroskedasticity (GARCH, Bollerslev, 1986) model. Theoretical and statistical relations are illustrated in the Appendix. Lamoureux and Lastrapes (1990, 1994) employ GARCH model to explain the time-varying relationships

¹ Whether cryptocurrency markets are efficient or not remains a controversial issue. Urquhart (2016) find that bitcoin market is not efficient, while Wei (2018) argues that liquidity must be considered. Note that Eugene Fama (supports EMH) and Robert Shiller (finds inefficient markets due to violation of variance bound requirements) received the Nobel Prize of Economic Sciences in the same year of 2013, indicating that further research is needed.

between information flow and volatility of stocks. They show the persistence of volatility can be captured by daily trading volume, which serves as a proxy for the rate of information flow in stock markets. It indicates volume acts as a subordinator, creating fat tails and volatility clustering as the ARCH (Engle, 1982) and GARCH models. In other words, the autocorrelation in the time-varying rate of information arrival causes volatility dependency. However, previous studies about GARCH and subordinated distribution with the Gaussian assumption cannot totally capture the extreme impact of information on volatility.

This study aims to explain the time-varying volatility and fat tailed distribution of cryptocurrencies by information flow dynamics. It investigates whether the unconditional fat-tailed return distribution is due to volatility persistence created by random arrival of information. Using the stable²-GARCH models to allow for heteroskedasticity and extreme impact reflected by fat tails, it examines how information influence the return and volatility of cryptocurrency portfolios from 1 January 2019 to 1 January 2025. Our results show that information flow, proxied by trading volume, significantly and positively affects volatility, while negatively impacts the probability of extreme returns. In addition, its impact on the tail behavior of return distribution varies with the size of trading volume. Furthermore, we find that the log-stable is more appropriate than log-normal for describing the price changes in both trading days and calendar days.

Several key contributions are made by this paper. It is the first to explain the fat-tailed distribution of cryptocurrency by information flow. We apply stable-GARCH model to capture the extreme impact of information on volatility. Second, the study documents log-stable distribution after considering the random arrival of information and shows fat-tailed distribution in both calendar time and trading time. It resolves the conflicts between log-normal by Clark (1973) and log-stable by Mandelbrot and Taylor (1967) in crypto markets. Third, the study highlights the tail risk of cryptocurrencies, which has important implications for portfolio managers, researchers and regulators in portfolio diversification, asset pricing and risk management.

² The other popular distributions for modelling financial time series are not considered because they do not meet the Central Limit Theorem. Moreover, the likelihood ratio tests of Hansen's (1994) flexible model that allows mean and variance to change, rejects the conditional t-distribution. Generalized Error distribution does not provide good fit either.

The rest of this paper is organized as follows. In the next section, we will provide the data and research methods. Empirical results will be reported and analyzed in Section 3. Conclusion and suggestions will be available in Section 4.

2. Data and Methodology

2.1 Data

There are more than 15 million cryptos in the market but only 248 cryptos are actively traded over six years with non-zero market capitalization. To ensure sufficient cross-sectional coverage of cryptocurrencies and a sample period covering the COVID-19 market wipeout, our sample comprises all 248 cryptocurrencies with complete time series from 1 January 2019 to 1 January 2025. The closing prices, trading volumes, and market capitalization at daily intervals are retrieved from CoinMarketCap, consisting of 1,635,930 observations. Given that higher trading volumes attract more analyst coverage creating more information flow (Damodaran, 1985), we sort cryptocurrencies into five quintiles based on daily trading volume, to compare information influence on return and volatility. Five value-weighted portfolios are constructed. The daily log-returns of the portfolio r_t at time t are calculated from their daily closing prices P_t :

$$r_t = \log P_t - \log P_{t-1} = \log\left(\frac{P_t}{P_{t-1}}\right) \quad (1)$$

2.2 Stable distribution

The stable distribution proposed by Mandelbrot (1963) and Fama (1965a) is used to depict the heavy tailed crypto returns. The characteristic function $X \sim S(\alpha, \beta, \gamma, \delta)$ can be described as in Nolan (2013), McCulloch (1996, 1997) and So (1978a, 1987b):

$$E[\exp(iuX)] = \begin{cases} \exp\left(-\gamma^\alpha |u|^\alpha \left[1 - i\beta \left(\tan \frac{\pi\alpha}{2}\right) (\text{sign}(\mu))\right] + i\delta u\right) & \alpha \neq 1 \\ \exp\left(-\gamma |u| \left[1 + i\beta \frac{2}{\pi} (\text{sign}(u)) \times \log(|u|)\right] + i\delta u\right) & \alpha = 1 \end{cases} \quad (2)$$

where $i = \sqrt{-1}$ is the imagine number. $\alpha \in (0, 2]$ denotes the characteristic exponent characterizing the heaviness of the tails. When $\alpha = 2$, it corresponds to normal distribution. The smaller the value of α , the heavier the tails. $\beta \in [-1, 1]$ describes skewness. $\beta = 0$ represents symmetric distribution. α and β are the shape parameters depicting *long-term*

tail risk. $\gamma > 0$ is the scale parameter measuring dispersion that is *short-term volatility*. $\delta \in \mathbb{R}$ is the location parameter. Because information flow within trading days may be different, our log-stable specification enables us to examine if the return and volatility patterns are influenced by the arrival of different types of information³.

2.3 Stable-GARCH

The conditional mean of crypto returns r_t is modeled by:

$$r_t = \mu + \varepsilon_t \quad (3)$$

Where μ is the constant term, and $\varepsilon_t \sim S(\alpha, \beta, \gamma, \delta)$ is the stable distributed residual term with shape parameter α , skewness β , scale γ , and location δ . The conditional variance h_t at time t in stable($\alpha, \beta, \gamma, \delta$)-GARCH (p, q) (hereafter stable-GARCH) process is described by Mittnik et al. (2002) as⁴:

$$h_t^\eta = \varpi + \sum_{i=1}^q \lambda_i |\varepsilon_{t-i}|^\eta + \sum_{j=1}^p \theta_j h_{t-j}^\eta \quad (4)$$

Where ϖ is a constant term representing the baseline volatility, parameter λ_i capture the ARCH effect (short-term volatility spikes), θ_j measures the GARCH effect (intermediate volatility persistence). $\varpi > 0$ and the coefficients λ_i ($i=0, \dots, p$) and θ_j ($j=1, \dots, q$) are all non-negative. The power parameter, η , satisfies $0 < \eta < \alpha$. The stable-GARCH process has a unique, strictly stationary solution if:

$$\kappa_{\alpha, \beta, \eta} \sum_{i=1}^q \lambda_i + \sum_{j=1}^p \theta_j \leq 1 \quad (5)$$

Where $\kappa_{\alpha, \beta, \eta} := E|\varepsilon_t|^\eta$. When $\eta = 1$, equation (5) reduces to:

$$\kappa_{\alpha, \beta, 1} = \frac{2}{\pi} \Gamma\left(1 - \frac{1}{\alpha}\right) (1 + \tau_{\alpha, \beta}^2)^{1/2\alpha} \cos\left(\frac{1}{\alpha} \arctan \tau_{\alpha, \beta}\right) \quad (6)$$

When volume is added to explain the volatility as in Lamoureux and Lastrapes (1990, 1994), equation (4) takes the form as:

³ These distinctions are important because scheduled macro-economic announcements, micro-economic information and option expiration may have different long-term or short-term impacts on volatility and trading volume, as documented by Bollerslev and Jubinski (1999) and Lienesfeld (2001).

⁴ Also see Francq and Meintanis (2016), Liu and Brorsen (1995), Mittnik et al. (2002), Panorska et al. (1995), Paoletta (2016).

$$h_t^\eta = \varpi + \sum_{i=1}^q \lambda_i |\varepsilon_{i,t-i}|^\eta + \sum_{j=1}^p \theta_j h_{j,t-j}^\eta + \xi v_{t-1} \quad (7)$$

Where v_{t-1} denotes the logarithm of volume and ξ is the parameter representing volume impacts. Parameters of stable-GARCH are estimated jointly using regression method based on the empirical characteristic function with an iterative algorithm to improve accuracy, according to Koutrouvelis (1980, 1981), which is less sensitive to outliers than maximum likelihood estimation.

3. Empirical results

Table 1 shows the daily return and volatility of crypto portfolios including stable distribution parameters estimates of returns. It documents the heavy tails of unconditional return distribution in calendar days (kurtosis > 3 and $\alpha < 2$). The Shapiro-Wilk (SW) test confirms the rejection of normality. The kurtosis of Portfolio 5 is much higher than the other portfolios. The size (market capitalization) of portfolios decreases and tails tend to be heavier from Portfolio 1 to 5 as volumes become lower. It indicates smaller cryptos have higher probability of realizing extreme returns than the larger cryptos. The Box-Ljung Q-statistics show both the daily return and log-volume series are serially correlated.

Table 1. The Return and Volatility of Crypto Portfolios Ranked by Volumes

	1	2	3	4	5
	HIGH				LOW
Min (r_t)	-0.478	-0.554	-0.512	-0.496	-0.547
Mean (r_t)	0.002	0.002	0.003	0.006	0.012
Median (r_t)	0.002	0.004	0.005	0.004	0.006
Max (r_t)	0.169	0.284	0.245	0.564	1.558
Std (r_t)	0.036	0.042	0.039	0.052	0.083
Skew (r_t)	-1.391	-1.857	-1.558	0.831	7.424
Kurt (r_t)	21.146	23.895	23.040	22.300	121.282
$\alpha(r_t)$	1.543	1.631	1.609	1.460	1.383
$\beta(r_t)$	-0.047	-0.322	-0.172	0.040	0.179
$\gamma(r_t)$	0.017	0.021	0.019	0.021	0.022
$\delta(r_t)$	0.002	0.006	0.006	0.004	0.005
E ($ r_t $)	0.024	0.028	0.026	0.032	0.038
E (VOL_t)	22,626,232,738	15,055,346	2,871,718	341,958	66,682
E (CAP_t)	431,027,491,537	503,872,278	260,532,461	64,081,442	22,755,897
SW (r_t)	0.898	0.878	0.881	0.821	0.542
Q ₂₀ (r_t)	42.444	54.427	70.049	90.115	247.18
Q ₂₀ (v_t)	23515	29883	24421	29055	28143

$Q_{20}(\Delta VOL_t)$	1436.7	355.5	479.99	456.3	646.13
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The portfolios are ranked in descending order based on daily trading volumes. r_t is the logarithm of portfolio's daily returns. v_t is the logarithm of portfolio daily trading volume. $SW(r_t)$ denotes the Shapiro-Wilk test statistics of crypto portfolio daily log-returns. $E(|r_t|)$ stands for the average of the absolute value of daily return of each portfolio. $E(VOL_t)$ is the average daily trading volume of each portfolio. $E(CAP_t)$ is the average daily market capitalization of each portfolio. $Q_{20}(r_t)$, $Q_{20}(v_t)$ and $Q_{20}(\Delta VOL_t)$ represent the Box-Ljung Q-statistics of portfolio's daily log-returns, log-volumes and log-changes of volumes, respectively.

The correlations of return, volatility and volume are presented in Table 2. Crypto returns are negatively correlated with volatility in Portfolio 1, 2 & 3 (larger-size cryptos), while the correlations are positive in Portfolio 4 & 5 (smaller-size cryptos). Volatility and volume are positively correlated, and the correlation decreases from 0.347 to 0.055 from Portfolio 1 to 5, suggesting higher volumes is associated with higher volatility. However, returns of are positively correlated with volume in Portfolio 2 & 3 (even though $\rho(r_t, |r_t|) < 0$ and $\rho(|r_t|, VOL_t) > 0$). In addition, correlations between return and log-volume v_t are positive, except for Portfolio 5.

Table 2. Pearson Correlation Coefficient for Return, Volatility and Volume

Panel A. Portfolio 1						Panel B. Portfolio 2					
	r_t	$ r_t $	VOL_t	v_t	ΔVOL_t		r_t	$ r_t $	VOL_t	v_t	ΔVOL_t
r_t	1.000					r_t	1.000				
$ r_t $	-0.126	1.000				$ r_t $	-0.245	1.000			
VOL_t	-0.012	0.347	1.000			VOL_t	0.041	0.216	1.000		
v_t	0.004	0.356	0.883	1.000		v_t	0.044	0.257	0.834	1.000	
ΔVOL_t	-0.013	0.342	0.265	0.243	1.000	ΔVOL_t	0.042	0.133	0.226	0.214	1.000
Panel C. Portfolio 3						Panel D. Portfolio 4					
	r_t	$ r_t $	VOL_t	v_t	ΔVOL_t		r_t	$ r_t $	VOL_t	v_t	ΔVOL_t
r_t	1.000					r_t	1.000				
$ r_t $	-0.147	1.000				$ r_t $	0.227	1.000			
VOL_t	0.104	0.193	1.000			VOL_t	0.109	0.104	1.000		
v_t	0.083	0.241	0.708	1.000		v_t	0.039	-0.040	0.690	1.000	
ΔVOL_t	0.119	0.160	0.356	0.306	1.000	ΔVOL_t	0.027	0.096	0.291	0.271	1.000
Panel E. Portfolio 5											
	r_t	$ r_t $	VOL_t	v_t	ΔVOL_t						
r_t	1.000										
$ r_t $	0.672	1.000									
VOL_t	0.023	0.055	1.000								
v_t	-0.046	-0.032	0.244	1.000							
ΔVOL_t	0.027	0.070	0.292	0.303	1.000						

r_t denotes the logarithm of portfolio's daily returns. $|r_t|$ denotes absolute return of the crypto portfolio, which serves as a measure of volatility. Absolute return, unlike standard deviation, focuses on the actual magnitude of price changes, offering a clearer picture of potential gains and losses, and is less susceptible to the influence of extreme outliers (Bandi and Russell, 2008). VOL_t denotes the trading volume of the crypto portfolio. v_t is the logarithm of portfolio daily trading volume. ΔVOL_t represents the daily log changes of the trading volume of the crypto portfolio.

Table 3 compares the shape of distribution between crypto returns and innovations in Equation (3). The characteristic exponents of returns and innovations are similar and less than 2 in all portfolios, showing heavy-tail distribution is needed for modelling the process. The left tails of the innovation distributions are displayed in Figure 2. With the lowest trading volumes, Portfolio 5 has the fattest tail among all crypto portfolios. Portfolio 1-3 have relatively lower probability of realizing extreme loss (e.g. beyond 80% loss).

Table 3: Stable Distribution Parameter Estimates of Crypto Returns and Innovations

	1		2		3		4		5	
Panel A. Crypto Portfolio Returns r_t										
α	1.543	***	1.631	***	1.609	***	1.460	***	1.383	***
	(-653.15)		(537.84)		(547.92)		(781.99)		(903.74)	
β	-0.047	***	-0.322	***	-0.172	***	0.040	***	0.179	***
	(-32.00)		(-195.91)		(-105.82)		(30.62)		(153.58)	
γ	0.017		0.021		0.019		0.021		0.022	
δ	0.002		0.006		0.006		0.004		0.005	
Panel B. Crypto Portfolio Innovations ε_t										
α	1.543	***	1.631	***	1.609	***	1.460	***	1.383	***
	(653.13)		(537.84)		(547.93)		(782.02)		(903.77)	
β	-4.71E-02	***	-0.322	***	-0.172	***	0.040	***	0.179	***
	(-32.01)		(-195.91)		(-105.83)		(30.62)		(153.58)	
γ	0.017		0.021		0.019		0.021		0.022	
δ	0.001		0.004		0.002		-0.002		-0.008	

The t-statistics are presented in parentheses. To examine whether the distribution is significantly different from normal distribution, the null hypothesis for alpha is: $\alpha < 2$; to determine whether the distribution is asymmetric, the null hypothesis for beta is: $\beta = 0$. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

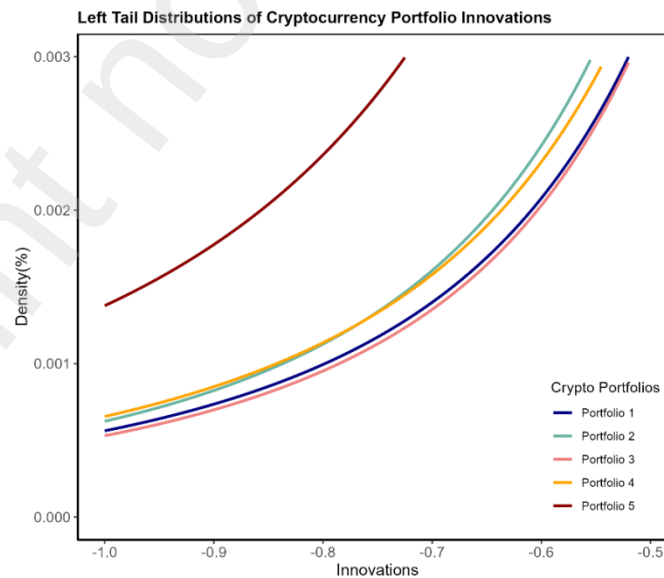


Fig. 2. Left Tail Distributions of Cryptocurrency Portfolio Innovations

Table 4 shows that ARCH and GARCH effects are significant among all portfolios under both the normal and stable assumption of innovations. But in normal-GARCH (Table 4(A)), the sum of persistence coefficients in Portfolios 3 & 4 is close to 1 ($\lambda + \theta = 0.99$) as in integrated-GARCH, which has infinite variance and is non-stationary. The amplified log-likelihood of stable-GARCH across all portfolios reaffirms it is more appropriate than normal-GARCH. In Table 4(B), cryptocurrencies exhibit significant volatility persistence ($\theta \approx 0.9$) in all portfolios. Portfolio 1 has the most prominent volatility persistence ($\theta = 0.928$). The ARCH effects are significant among all portfolios while lower ($\lambda < 0.1$) than the GARCH effects. The constant terms ϖ are significant for all portfolios, indicating inefficient market. After considering conditional heteroskedasticity, the conditional distribution become closer to normal distribution in all portfolios yet remains heavy tailed.

Table 4. Parameter Estimates of GARCH(1,1) Model without Volume

	1	2	3	4	5
Panel A: Normal-GARCH(1,1) Model					
ϖ	0.00007 ** (2.49)	0.00009 * (1.78)	3.10E-05 (1.41)	0.00007 * (1.66)	0.00019 (0.9)
λ	0.109 ** (2.27)	0.154 *** (3.27)	0.098 *** (2.92)	0.171 ** (2.49)	0.183 *** (3.26)
θ	0.845 *** (18.43)	0.814 *** (15.82)	0.892 *** (27.86)	0.828 *** (13.58)	0.816 *** (14.73)
LLH	4325.61	4048.06	4205.17	3776.83	3015.12
Panel B: Stable-GARCH(1,1) Model					
ϖ	0.0003 *** (89.02)	0.0010 *** (95.49)	0.0007 *** (92.62)	0.0004 *** (83.49)	0.0006 *** (111.09)
λ	0.044 *** (249.31)	0.079 *** (193.58)	0.064 *** (180.92)	0.051 *** (191.38)	0.034 *** (189.53)
θ	0.928 *** (2822.55)	0.858 *** (937.67)	0.886 *** (1161.75)	0.917 *** (1768.43)	0.924 *** (2025.58)
α	1.614 *** (491.03)	1.754 *** (349.36)	1.721 *** (382.16)	1.607 *** (561.98)	1.531 *** (684.43)
β	-0.009 *** (-11.72)	-0.361 *** (-166.67)	-0.215 *** (-118.42)	-0.088 *** (-66.18)	-0.052 *** (-37.94)
LLH	4563.23	4277.81	4436.71	4087.88	3742.55

The t-statistics in parentheses are measured with robust standard errors. LLH represents the Log-likelihood of the model. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

Table 5 shows the impact of information flow on the estimates of GARCH models. In stable-GARCH (Table 5(B)), volume is significant across all portfolios, suggesting information flow act as a significant subordinator directing the return generating process

of cryptocurrency. Volume is positively associated with the volatility of unexpected price movement (h_t). The GARCH and ARCH effects are still significant, but the level of the impacts decrease after the inclusion of volume. Its impacts are different for large and small cryptos: Portfolios 2 and 3 present the largest volume impact and lightest tails. Their ϖ become insignificant, indicating volume captures the abnormal return. Portfolio 1 demonstrates the lowest volume impact. It has a larger portion of liquidity trading and is consistent with Li and Wu (2006) who have shown that liquidity trading could reduce volatility. Trading volumes affect the kurtosis and tail exponents of returns in large and small crypto portfolios differently, which aligns with Westerfield (1977) in stock markets. The results show that volume reflects an uneven but persistent information flow and have significant impacts on the nonnormality of unconditional return distribution. Figure 3 shows the behavior of GARCH and volume across time. The impacts of Covid-19 are discussed in Appendix B.

Table 5. Parameter Estimates of GARCH(1,1) Model with Volume

	1		2		3		4		5
Panel A: Normal-GARCH(1,1) Model									
ϖ	1.30E-08		8.60E-11		2.49E-10		1.38E-09		0.00004
	(0.00)		(0.00)		(0.00)		(0.00)		(0.47)
λ	0.108	***	0.155	***	0.098	***	0.188	***	0.319
	(76.57)		(235.65)		(40.35)		(99.48)		(9.29)
θ	0.840	***	0.803	***	0.888	***	0.828	***	0.820
	(118.65)		(58.28)		(151.63)		(99.41)		(28.73)
ξ	0.000003	***	0.000006	***	0.000002	***	0.000005	***	7.87E-13
	(161.09)		(3654.72)		(94.52)		(317.87)		(0.00)
LLH	4325.45		4049.99		4204.56		3778.37		3055.70
Panel B: Stable-GARCH(1,1) Model									
ϖ	5.04E-09	***	5.33E-11		2.56E-10		0.00005	***	0.00010
	(16.49)		(0.19)		(0.001)		(22.52)		(8.72)
λ	0.044	***	0.081	***	0.067	***	0.051	***	0.033
	(251.17)		(198.49)		(279.86)		(191.67)		(172.25)
θ	0.926	***	0.840	***	0.873	***	0.916	***	0.924
	(2775.11)		(809.47)		(2069.7)		(1764.33)		(1628.78)
ξ	0.00001	***	0.00008	***	0.00006	***	0.00003	***	0.00006
	(97.2)		(99.91)		(321.65)		(69.13)		(75.41)
α	1.615	***	1.757	***	1.727	***	1.608	***	1.534
	(489.36)		(345.48)		(48.36)		(561.82)		(680.47)
β	-0.009	***	-0.368	***	-0.224	***	-0.088	***	-0.051
	(-15.15)		(-167.15)		(-630.81)		(-65.74)		(-26.86)
LLH	4564.03		4281.17		4439.14		4088.05		3745.71

The t-statistics in parentheses are measured with robust standard errors. LLH represents the Log-likelihood of the model. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

In Table 6, after accounting for the random arrival of information within trading days, the standardized innovations become less heavy tailed compared to the results in Table 3. The shape parameter α significantly increases across all crypto portfolios. Having the largest volume impacts, Portfolios 2 & 3 show the lightest tails of standardized innovations among all portfolios. Their α increase from 1.6 (Table 3(B)) to 1.7 (Table 6(B)). However, the characteristic exponents are still significantly less than 2.0 after adjusting for heteroskedasticity. Thus, the log-stable recommended by Mandelbrot and Taylor (1967) is more appropriate than log-normal for describing the returns within trading days and calendar days⁵.

Table 6. Stable Distribution Parameters of Crypto Portfolios Return and Residual															
	1			2			3			4			5		
Panel A: Standardized Residuals in GARCH model ε_t/h_t															
α	1.608	***	1.721	***	1.706	***	1.608	***	1.503	***					
	(559.92)		(426.38)		(437.11)		(559.89)		(726.08)						
β	-0.048	***	-0.574	***	-0.369	***	-0.023	***	0.208	***					
	(-29.01)		(-318.05)		(-194.29)		(-14.17)		(152.97)						
γ	0.996		0.978		0.990		0.998		0.968						
δ	0.043		0.212		0.140		-0.067		-0.337						
Panel B: Standardized Residuals in GARCH model with Volume ε_t/h_t															
α	1.611	***	1.723	***	1.708	***	1.608	***	1.505	***					
	(545.62)		(424.84)		(427.05)		(559.88)		(724.88)						
β	-4.82E-02	***	-0.583	***	-0.377	***	-0.023	***	0.217	***					
	(-29.24)		(-322.02)		(-197.95)		(-13.91)		(158.73)						
γ	0.997		0.979		0.988		0.998		0.968						
δ	0.043		0.211		0.140		-0.068		-0.342						

The t-statistics are presented in parentheses. To examine whether the distribution is significantly different from normal distribution, the null hypothesis for alpha is: $\alpha < 2$; to determine whether the distribution is asymmetric, the null hypothesis for beta is: $\beta = 0$. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

⁵ Note that the α estimates are closely related to the fractal integration model recommended by Bollerslev and Jubinski (1999) in studying the long-term nature of volatility. The fractal integration model implies again that the Clark's (1973) log-normal assumption is not appropriate because, by definition, the fractal model will not be normal.

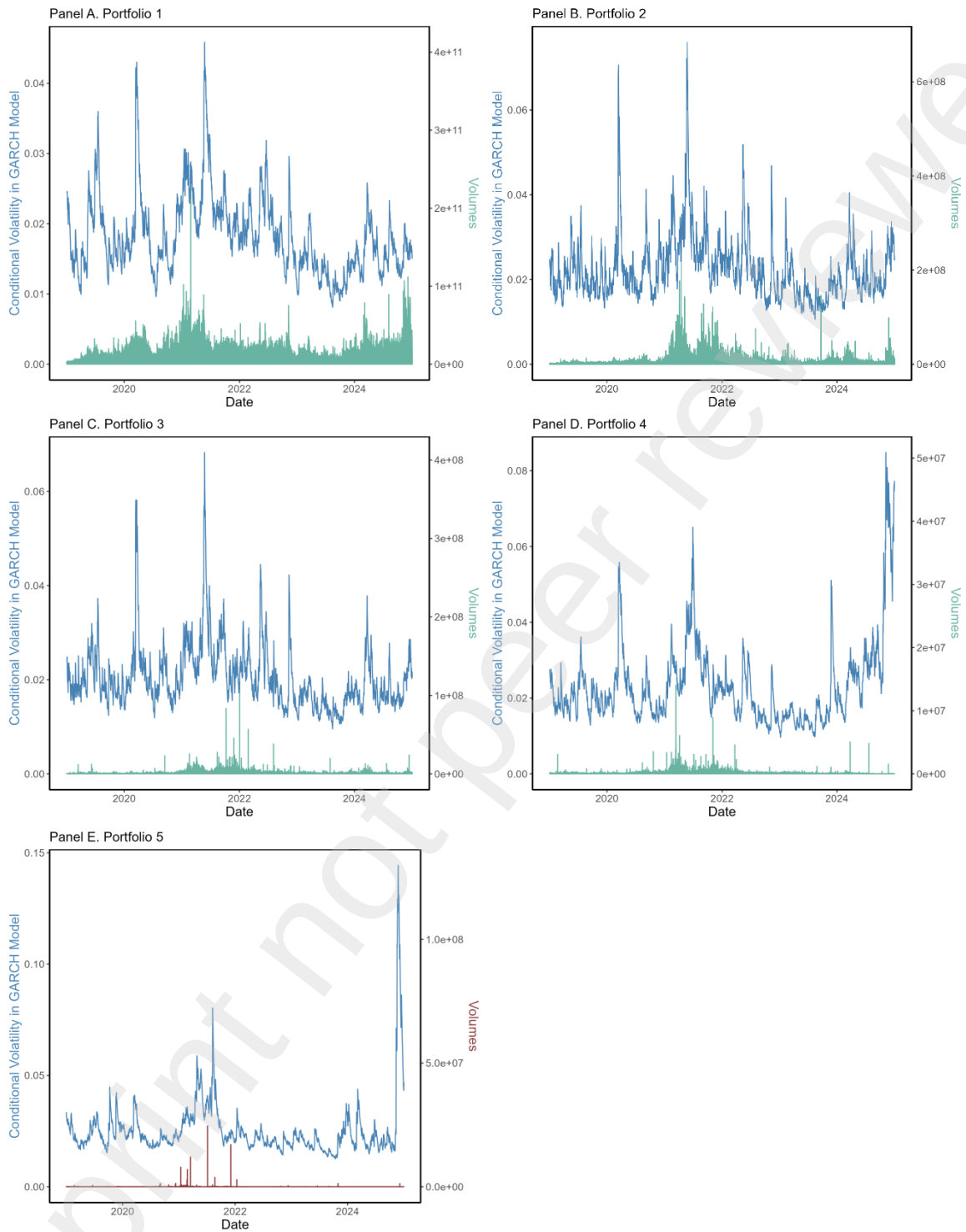


Fig. 3. Conditional Volatility in GARCH Model with Volume

Conclusion

This study finds that information flow, proxied by trading volumes, positively affects the time-varying volatility while negatively influences tail heaviness of crypto return distribution. Its impacts are different for large and small cryptos. Cryptocurrency with higher trading volumes exhibits lighter tails, indicating a smaller probability of realizing extreme loss. It is also found that after adjusting for heteroskedasticity created by arrival of information, crypto returns' characteristic exponents are still around 1.6 to 1.7, significantly less than 2.0. Thus, the log-normal assumption of the subordinated distribution by Clark (1973) is rejected, and the log-stable recommended by Mandelbrot and Taylor (1967) is more appropriate for describing the information flow within trading days and calendar days. Unlike Lamoureux and Lastrapes (1990,1994), this study verifies that the GARCH effect remains significant statistically after including trading volumes as proxy of latent information variables. Our tests are less biased because the information impacts on long-term volatility are captured by the characteristic exponents and thus decreases endogeneity between volatility and volume.

Researchers and regulators are recommended to consider “tail risk” in developing economic models to prevent extreme losses, especially during economic crises. The diversification effect will be lower with stable distribution compared to the Gaussian assumption (Fama, 1965a). In addition, the size effect and growth effect are the key factors of the Fama-French's (1992) factor model. By comparing crypto portfolios ranked by size, this study sheds light on asset pricing models for cryptocurrencies. Future research is encouraged to employ more nuanced measures of information flow to account for the noises of high-frequency trading and market microstructure, since volume could be contaminated by non-informed trading (Brogaard et al., 2014).

Appendix A.

The EMH states that the anticipated change in the price between today and tomorrow cannot be predicted by anything that influenced the past prices. The best forecast of tomorrow's price P_{t+1} is today's price P_t :

$$E(P_{t+1}|P_t, P_{t-1}, P_{t-2}, \dots) = P_t \quad (\text{A.1})$$

EMH only requires market to be a “martingale” or a “fair game”, that means information cannot be used to make arbitrage profits:

$$E(X_{t+1}|\Phi_t) = X_t \quad (\text{A.2})$$

where X_t is a martingale with respect to the information set Φ_t . On the other hand, a stochastic process Y_t is a fair game if it has the following property:

$$E(Y_{t+1}|\Phi_t) = 0 \quad (\text{A.3})$$

The subordinated normal model states that each price change measured over calendar time can be thought of as the sum of a random number, n , of identically distributed random variables, X (Westerfield, 1977). The return of P_t in equation (A.1) can be expressed as:

$$r_t = \mu_t + \sum_{i=1}^{n(t)} \mu_i \quad (\text{A.4})$$

where $\mu_t \sim i.i.d. N(0, \sigma^2)$ denotes forecastable component, μ_i are incremental changes. $n(t)$ is the number of new information in period t (within trading days), which is a serially dependent unobservable random variable and independent of μ_i .

Because of the randomness of $n(t)$, r_t is not normally distributed. It is a mixture of normal distributions. r_t is a subordinated stochastic process, where $r_t - \mu_t$ is subordinated to μ_i , and $n(t)$ is the directing process. Bera and Higgins (1993) has shown that equation (A.4) can be rewritten as:

$$r_t = \mu_t + \gamma n(t)^{1/2} \psi_t \quad (\text{A.5})$$

where ψ_t is standard normal distribution $N(0,1)$. Assuming that the information Φ_t include all current and historical information, r_t is conditional on the information set and the directing process, $n(t)$:

$$r_t | (\Phi_{t-1}, n(t)) \sim N(\mu_t, \gamma^2 n(t)) \quad (\text{A.6})$$

which is a conditional heteroskedastic normal distribution. Since $n(t)$ is not observable and if a random variable condition on $n(t)$ is normal, unconditionally, it must be non-normal as shown by the Paretian stable distribution. Moreover, the distribution for $r_t | \Phi_{t-1}$, would be conditionally heteroskedastic and non-normal. That is closely related to the GARCH by Bollerslev (1986), of which the conditional mean equation is:

$$r_t = \mu + \varepsilon_t \quad (\text{A.7})$$

Where μ is the constant, $\varepsilon_t = \sigma_t z_t$ and $z_t \sim N(0,1)$. The conditional variance h_t^2 is described as:

$$h_t^2 = \varpi + \sum_{i=1}^q \lambda_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \theta_j h_{t-j}^2 \quad (\text{A.8})$$

Where λ_i and θ_j are coefficients of the ARCH and GARCH terms.

Appendix B.

Figure 3 reveals that cryptocurrencies have dramatical conditional volatility, which is related to volume. Volatility varies during different time period. Thus, structural changes should be considered. It can also serve as a robustness test for Mandelbrot and Taylor's log-stable distribution. We therefore examine the returns with stable-GARCH during COVID-19 to assess the impact of volume during crisis. The parameter estimates are illustrated in Table B.7. During COVID-19, the impacts of volume are significant in all portfolios. For larger cryptos (Portfolio 1-4), the volume impacts magnify with higher ξ and lower θ compared to that in Table 5. For Portfolio 5, however, the volume effect reduces while volatility persistence increases compared to that in Table 5.

Table B.7. Parameter Estimates of Stable-GARCH(1,1) Model during Covid

Panel A: Stable-GARCH(1,1) Model without Volume															
	1			2			3			4			5		
ϖ	0.001	***		0.001	***		0.001	***		0.001	***		0.0004	***	
	(92.20)			(114.53)			(93.56)			(73.93)			(86.84)		
λ	0.049	***		0.085	***		0.082	***		0.051	***		0.036	***	
	(181.52)			(221.11)			(171.44)			(141.34)			(143.49)		
θ	0.912	***		0.840	***		0.847	***		0.906	***		0.935	***	
	(1755.17)			(975)			(820.33)			(1148.14)			(1970.72)		
α	1.674	***		1.785	***		1.721	***		1.607	***		1.647	***	
	(321.71)			(214.50)			(292.69)			(410.36)			(373.74)		
β	-0.068	***		-0.449	***		-0.218	***		-0.067	***		-0.189	***	
	(-28.91)			(-120.82)			(-85.64)			(-36.78)			(-94.35)		
LLH	2322.70			2173.57			2272.84			2277.16			2051.81		
Panel B: Stable-GARCH(1,1) Model with Volume															
ϖ	3.42E-09	***		2.31E-10	**		1.71E-09	***		1.36E-11			4.34E-10	*	
	(19.03)			(2.07)			(74.36)			(0.08)			(1.78)		
λ	0.049	***		0.084	***		0.083	***		0.051	***		0.034	***	
	(183.36)			(221.43)			(335.22)			(138.33)			(1494.3)		
θ	0.909	***		0.826	***		0.835	***		0.896	***		0.933	***	
	(1696.54)			(875.14)			(3950.48)			(1010.9)			(4327.17)		
ξ	0.00003	***		0.00011	***		0.00009	***		0.00006	***		0.00005	***	
	(93.92)			(119.8)			(598.74)			(77.88)			(114.02)		
α	1.676	***		1.789	***		1.729	***		1.611	***		1.652	***	
	(319.68)			(211.82)			(310.39)			(409.4)			(7626.85)		
β	-0.069	***		-0.453	***		-0.225	***		-0.062	***		-0.191	***	
	(-32.82)			(-121.86)			(-827.00)			(-34.16)			(-95.25)		
LLH	2323.15			2175.93			2274.60			2279.06			2053.26		

WHO declares COVID-19 as a public health emergency from 30 January 2020 to 5 May 2023. The t-statistics in parentheses are measured with robust standard errors. LLH represents the log-likelihood of the model. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

After COVID-19, the GARCH effect increases in Portfolio 1-4 (Table B.8) compared to that in Table B.7. In contrast, the volume effect reduces among all portfolios except that it remains the same level in the largest portfolio (Portfolio 1). Volumes become insignificant in Portfolio 4 & 5. With low trading volumes, the behavior of small size crypto portfolios is similar to “small firms” that may suffer from infrequent trading and thus biased systematic risk estimates (Banz, 1981; Roll, 1981; Dimson, 1979).

Table B.8. Parameter Estimates of Stable-GARCH(1,1) Model Post-Covid

Panel A: Stable-GARCH(1,1) Model without Volume															
	1			2			3			4			5		
ϖ	0.0001	***		0.0005	***		0.0004	***		0.0002	***		0.0010	***	
	(30.05)			(277.29)			(44.87)			(20.56)			(108.00)		
λ	0.027	***		0.069	***		0.054	***		0.058	***		0.038	***	
	(132.59)			(5160.93)			(119.71)			(66.44)			(138.82)		

θ	0.957 ***	0.890 ***	0.913 ***	0.922 ***	0.891 ***
	(2352.8)	(17564.87)	(916.21)	(658.15)	(1143.8)
α	1.557 ***	1.786 ***	1.792 ***	1.653 ***	1.410 ***
	(303.14)	(2322.72)	(119.03)	(244.39)	(481.89)
β	0.039 ***	-0.455 ***	-0.263 ***	-0.021 ***	0.177 ***
	(18.76)	(-689.42)	(-61.44)	(-4.00)	(70.84)
LLH	1423.29	1323.95	1380.98	1043.67	1036.48
Panel B: Stable-GARCH(1,1) Model with Volume					
ϖ	9.93E-11	1.63E-10 **	9.26E-13	0.0002 ***	0.0010 ***
	(0.61)	(2.13)	(0.00)	(23.25)	(108.37)
λ	0.027 ***	0.070 ***	0.055 ***	0.058 ***	0.038 ***
	(130.05)	(108.2)	(119.55)	(72.17)	(139.22)
θ	0.954 ***	0.880 ***	0.903 ***	0.922 ***	0.891 ***
	(2049.11)	(591.05)	(773.45)	(720.22)	(1148.24)
ξ	0.00001 ***	0.00004 ***	0.00004 ***	0.00000	0.00000
	(33.56)	(45.71)	(45.88)	(0.12)	(0.05)
α	1.558 ***	1.790 ***	1.797 ***	1.653 ***	1.410 ***
	(299.73)	(165.92)	(117.9)	(242.68)	(481.79)
β	0.038 ***	-0.472 ***	-0.276 ***	-0.021 ***	0.177 ***
	(18.10)	(-101.52)	(-63.39)	(-7.11)	(70.80)
LLH	1423.48	1324.58	1381.48	1043.67	1036.48

Starting from 5 May 2023, WHO declares end to COVID-19 as a public health emergency. The t-statistics in parentheses are measured with robust standard errors. LLH represents the log-likelihood of the model. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

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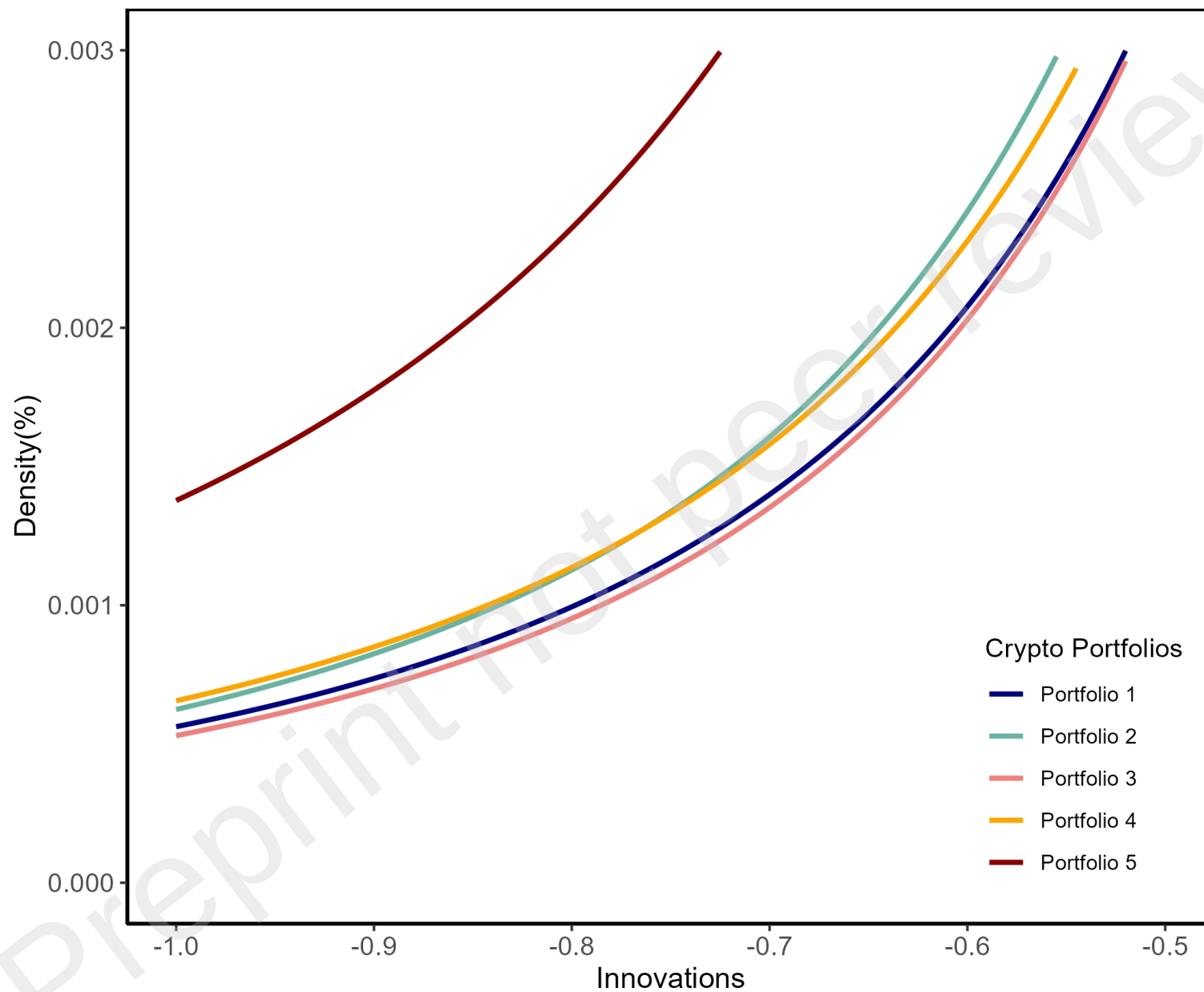
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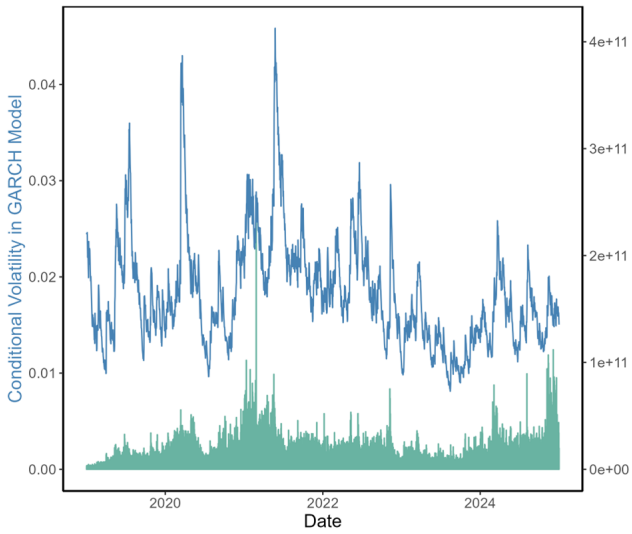
Price, Volatility and Volumes of Bitcoin (1.1.2015~1.1.2025)



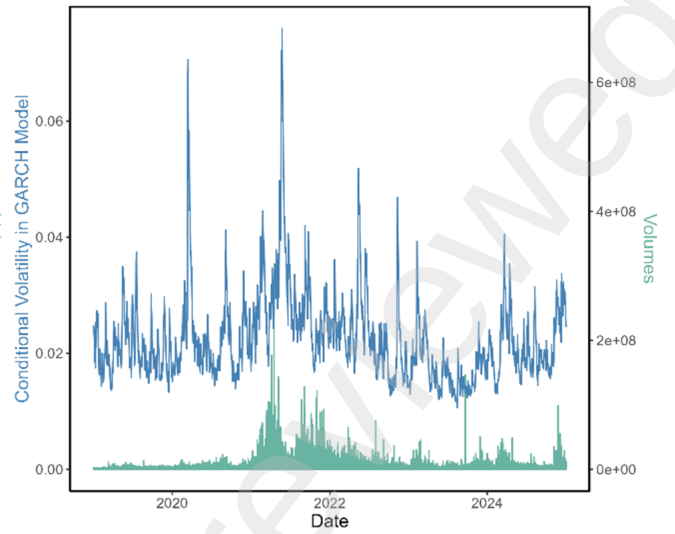
Left Tail Distributions of Cryptocurrency Portfolio Innovations



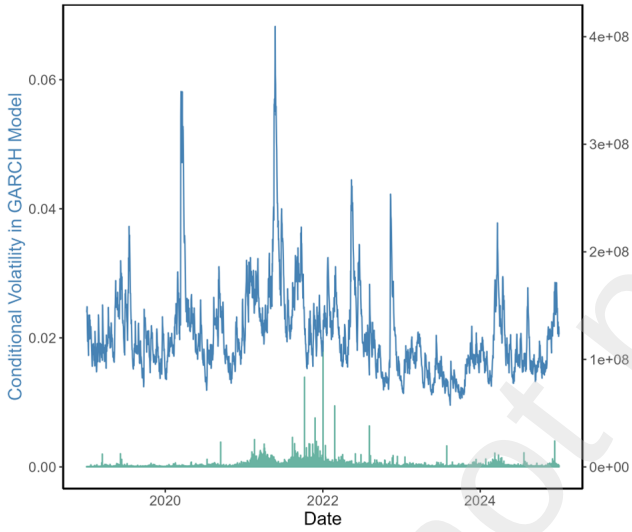
Panel A. Portfolio 1



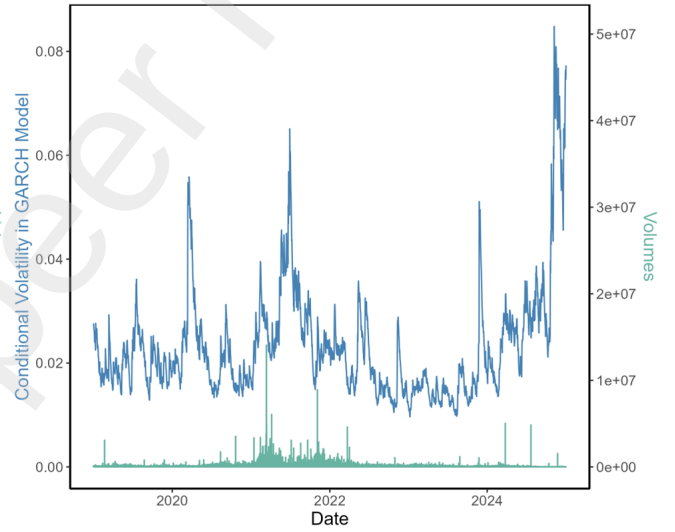
Panel B. Portfolio 2



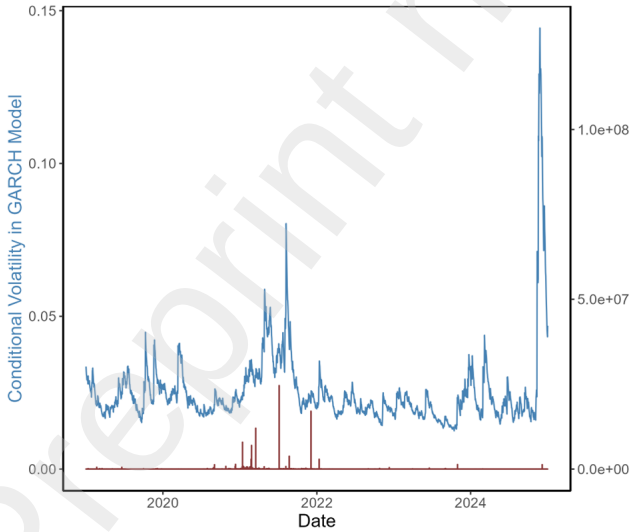
Panel C. Portfolio 3



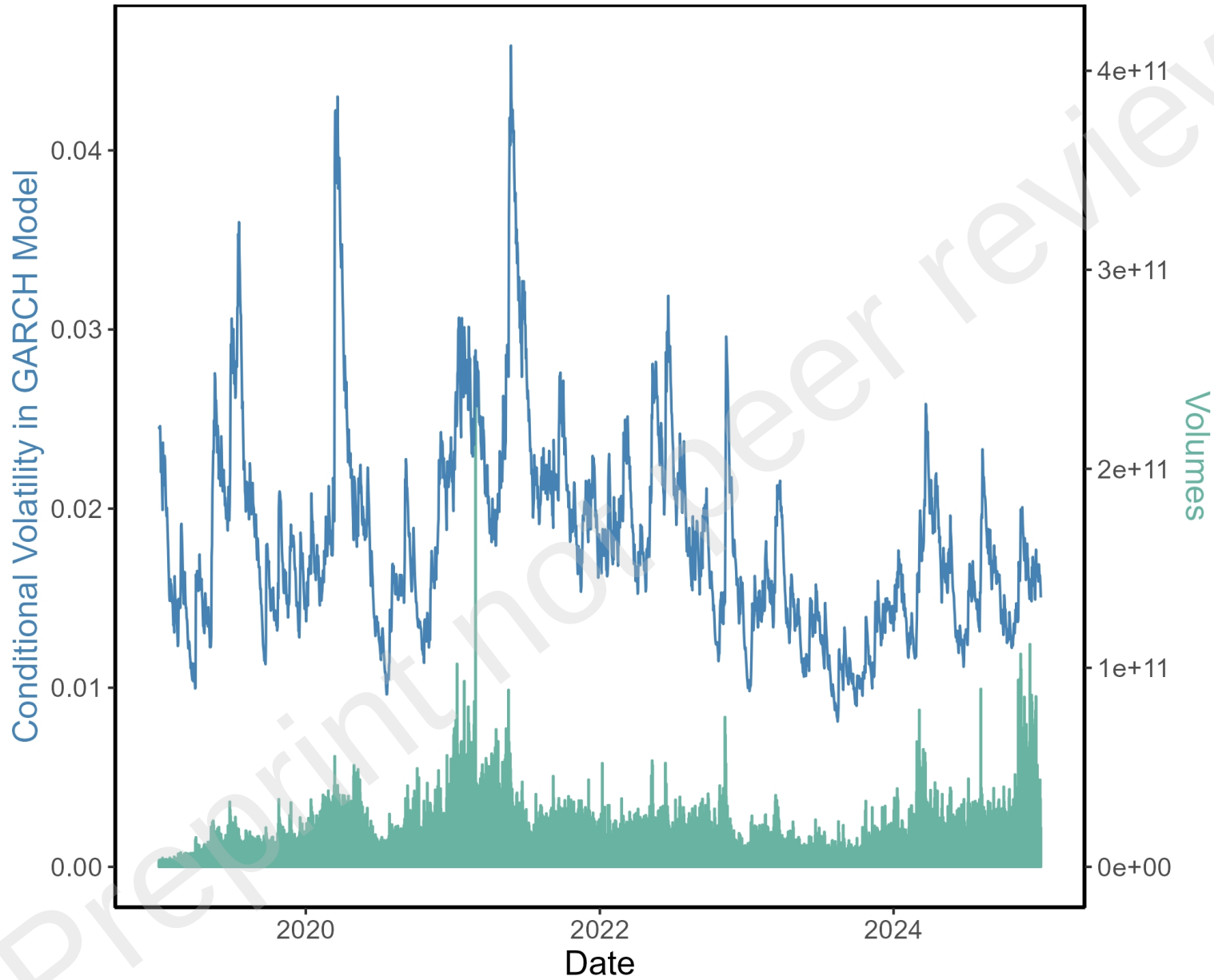
Panel D. Portfolio 4



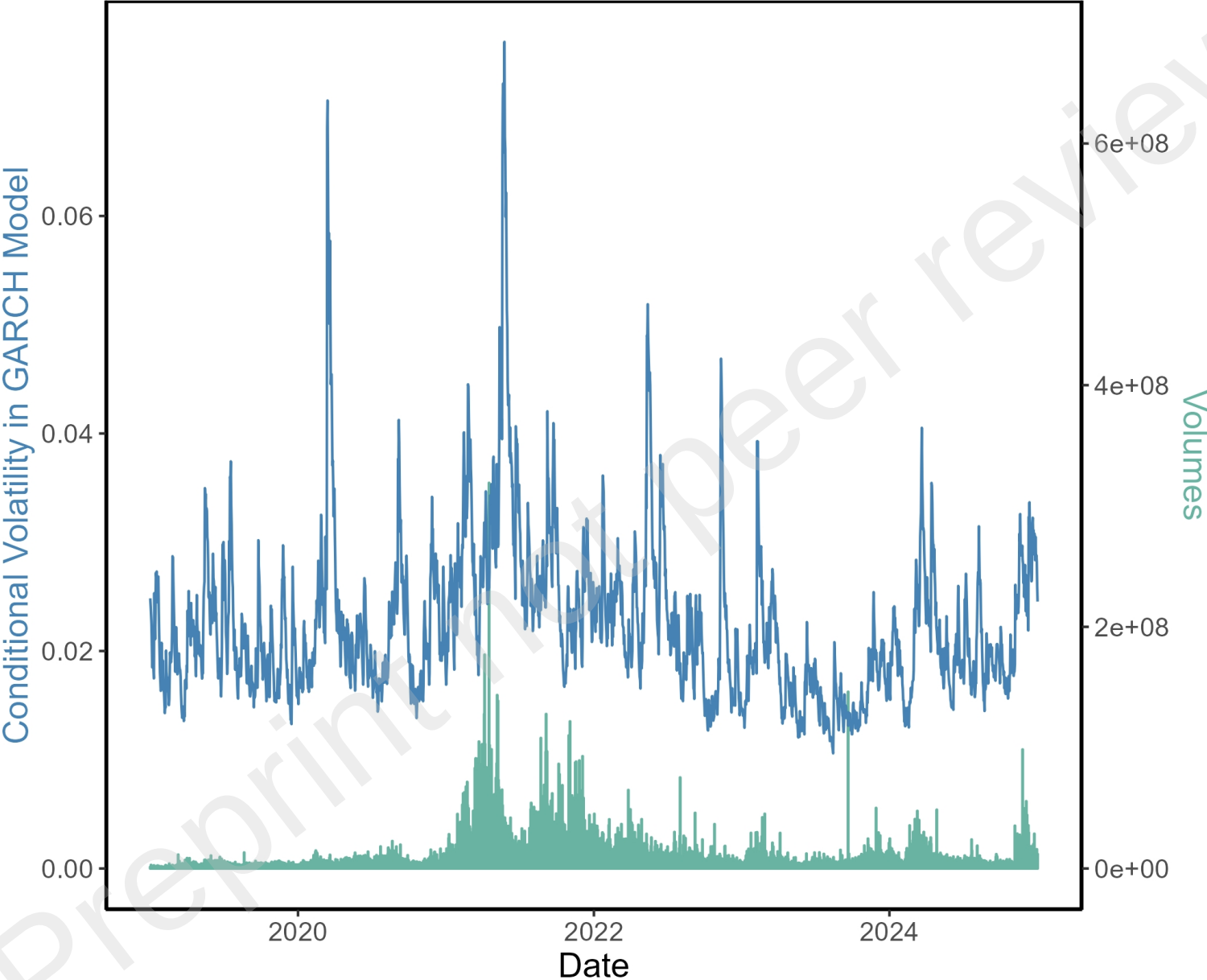
Panel E. Portfolio 5



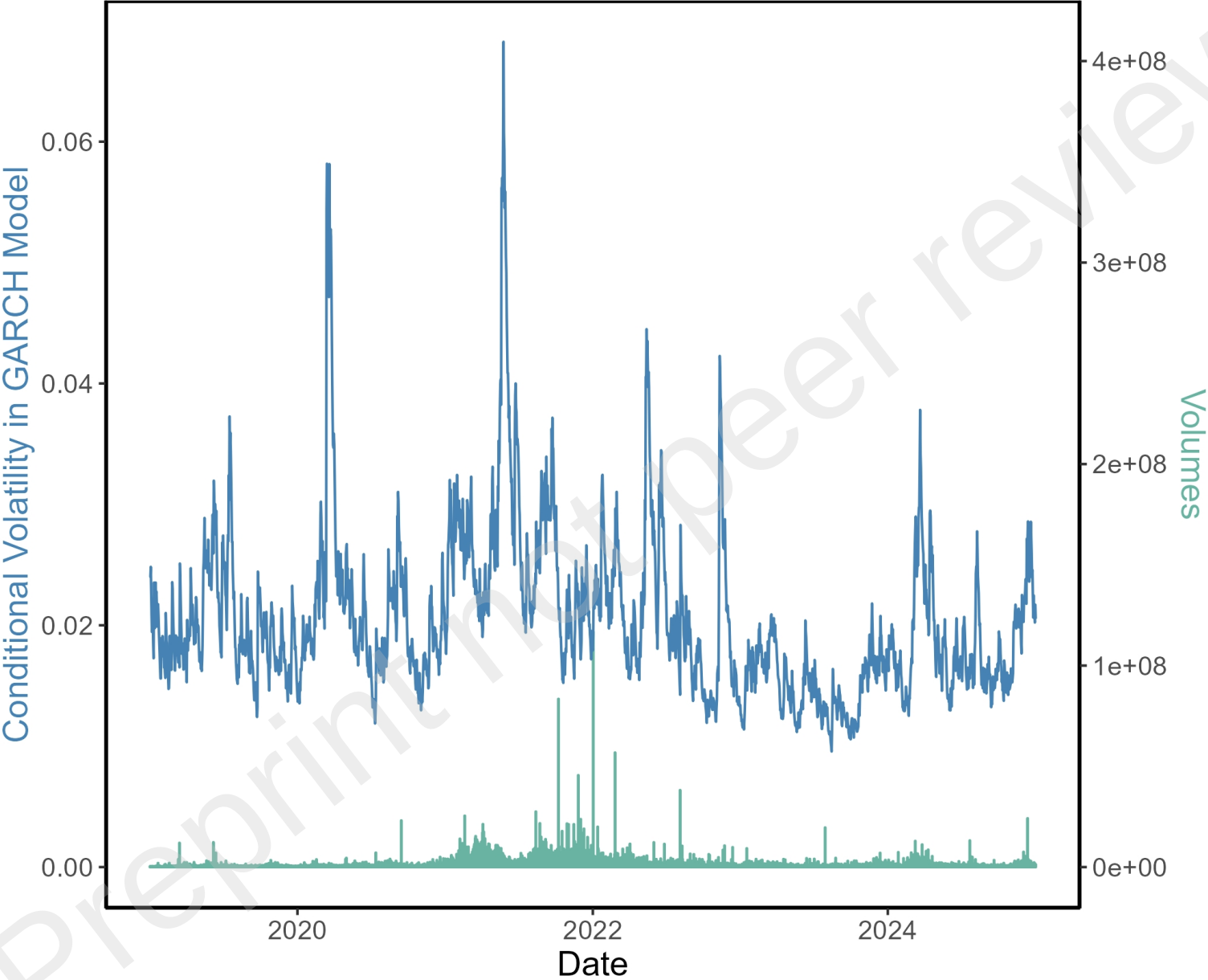
Panel A. Portfolio 1



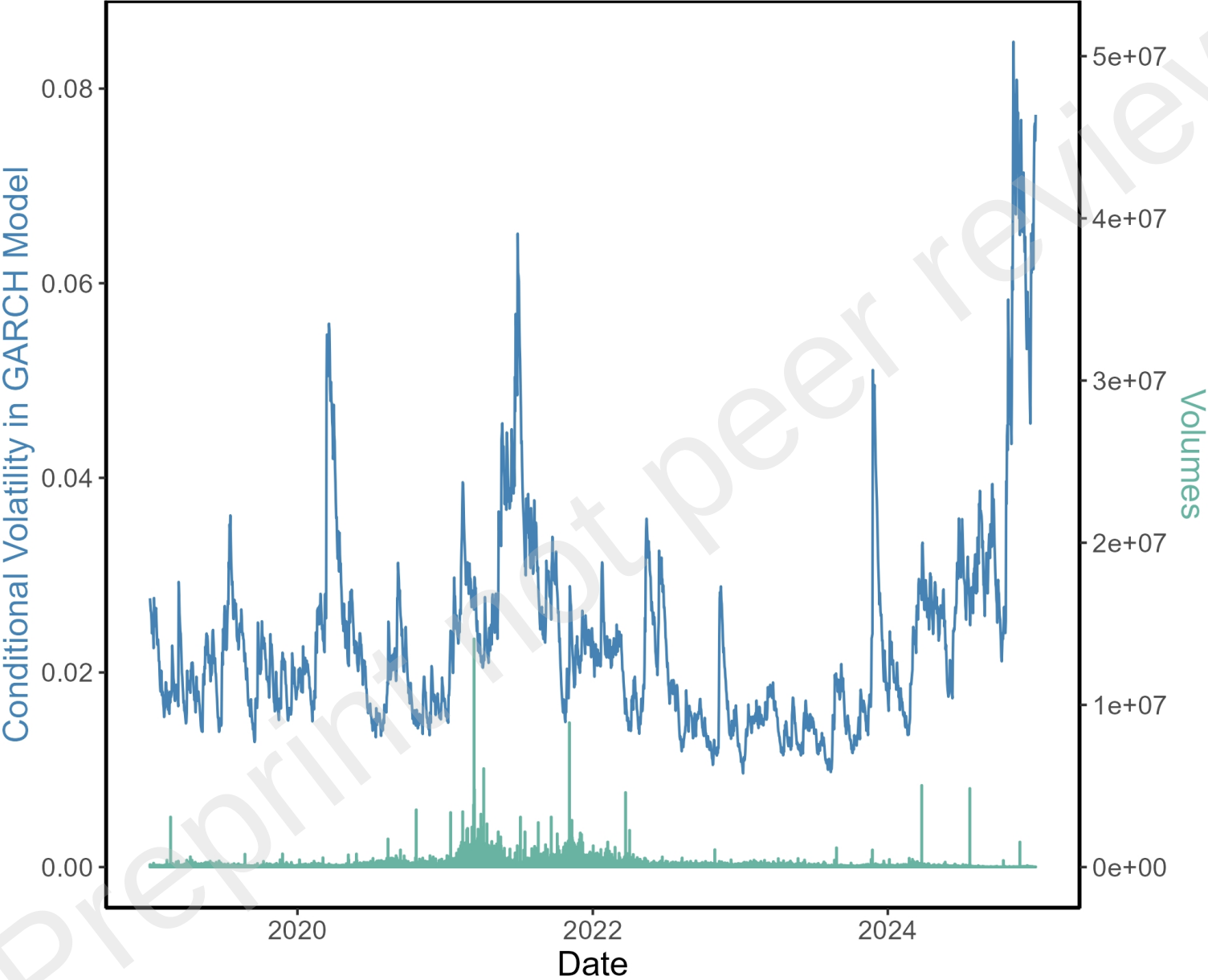
Panel B. Portfolio 2



Panel C. Portfolio 3



Panel D. Portfolio 4



Panel E. Portfolio 5

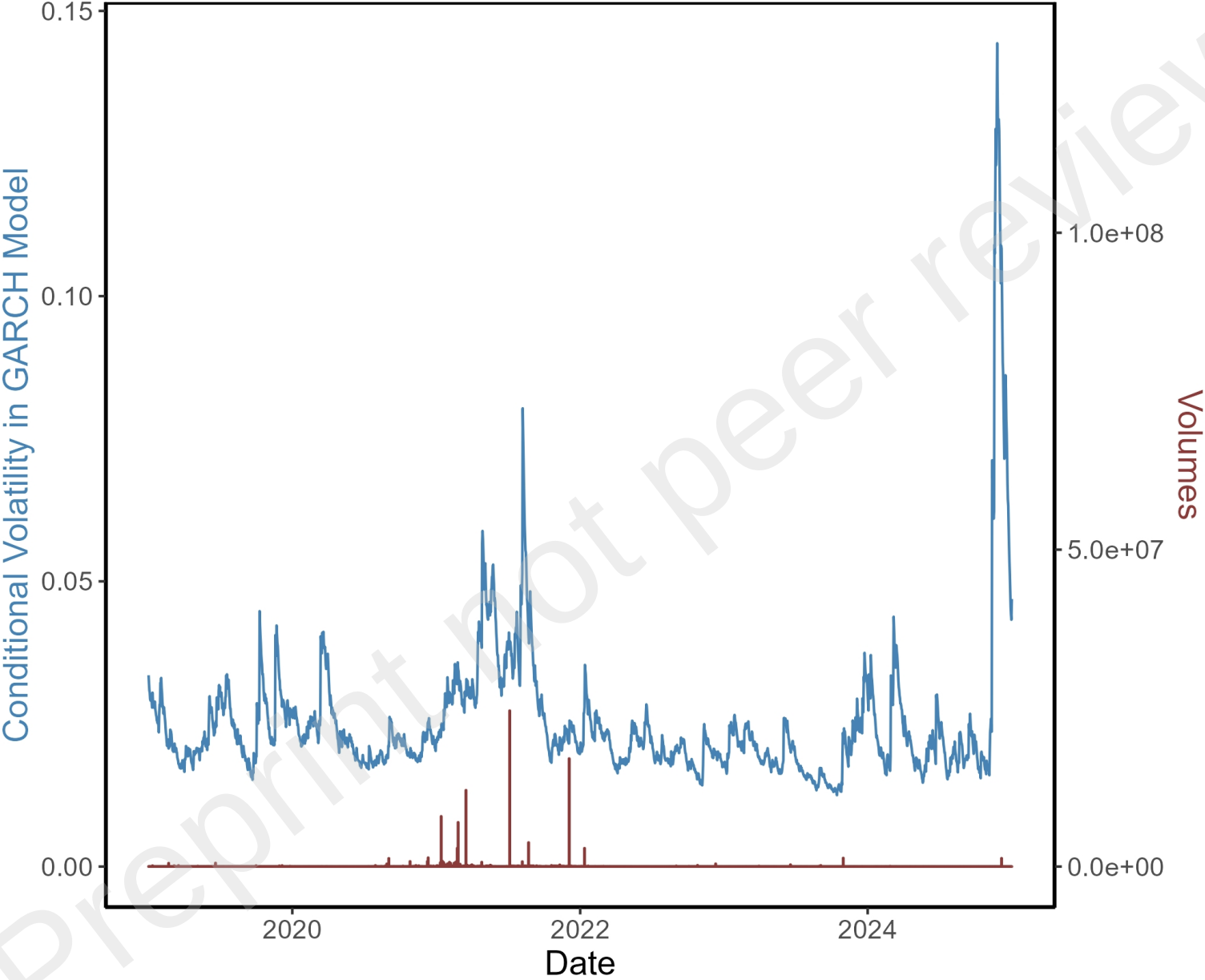


Figure Captions

Fig. 1. Price, Volatility and Volumes of Bitcoin

Fig. 2. Left Tail Distributions of Cryptocurrency Portfolio Innovations

Fig. 3. Conditional Volatility in GARCH Model with Volume